

PROJECT PROPOSAL

**WorkSure: A Service Marketplace Platform
for Connecting Customers with Verified Workers**

Course	Advanced Object Oriented Programming Language Lab	Section	A
Course Type	Laboratory / Group Project	Term	Academic Lab Project

Group Members

SL	Name	Student ID	Role in the Project
1	Al Araf Mahmud Samir	0112230624	Backend (Java/Spring), database, security
2	Marfatul Hossain	0112230701	Frontend (HTML/CSS/JavaScript), UI/UX, API wiring
3	Taofik Ahamed	0112230831	Bookings, reviews, testing & documentation

1. Introduction and Problem Statement

Finding a trusted electrician, cleaner, babysitter, or similar worker is still difficult for many people. Customers do not know who is verified, workers do not have a simple way to receive bookings, and there is often no clear record of price or job status. For our Advanced Object Oriented Programming Language Lab, we propose **WorkSure**, a web-based service marketplace that connects three types of users—customers, workers, and an admin—through one platform. The project is designed so that object-oriented ideas (classes, encapsulation, abstraction, layered design, and dependency injection) are applied in a real Java backend, not only in theory.

2. Objectives

- Build a complete lab project where Java classes work together as a server (API) and a plain HTML/CSS/JavaScript website acts as the user interface.
- Allow customers to browse six service sectors (Cleaning, Electrician, Security, Catering, Babysitting, Pet Care), book a worker and review completed jobs.
- Allow workers to create a profile, list services, accept or reject jobs, upload documents for verification, and read received reviews.
- Give an admin dashboard to verify workers, manage users, and view bookings and reviews.
- Practice OOP in Spring Boot: separate controllers, security, data access, and reusable helper classes, with JWT login and role-based access.

3. Proposed System (What We Will Build)

WorkSure will be a three-part system. The **frontend** is a plain HTML/CSS/JavaScript website (pages, forms, dashboards). The **backend** is a Java Spring Boot application that receives requests, checks who is logged in, and applies business rules. The **database** is MariaDB/MySQL, which stores users, services, bookings, reviews, and verification records. The website never talks to the database directly; every click that needs data goes through the Java server. Spring Boot will serve the HTML pages and API together at `http://localhost:5000`.

Main modules: (1) Authentication and roles, (2) Service catalog and worker listings, (3) Customer bookings and worker job transitions, (4) Worker profile and services, (5) Completed-booking reviews, (6) Private document verification, (7) Admin users and read-only reports.

4. Technology Stack and Why It Fits This Course

Java 17 and Spring Boot are chosen because this is an OOP lab: we can show classes, constructor injection, interfaces (for example password encoding), a security filter, and exception handling in Java. **Spring JDBC** (not JPA) will talk to **MariaDB** using clear SQL so table design and foreign keys stay visible for learning. **JWT + Spring Security + BCrypt** will protect routes (customer / worker / admin). **Plain HTML, CSS and JavaScript with browser fetch()** will provide

the interface so the Java API can be demonstrated as a full working product. Together, the stack matches a modern OOP lab: object design on the server, plus a complete end-to-end application.

5. Object-Oriented Concepts to Be Applied

- **Encapsulation:** hide secrets and messy details inside classes (JWT signing, file upload rules, password hashing).
- **Abstraction:** controllers call simple methods such as query/insert; they do not deal with raw JDBC code.
- **Composition and DI:** a booking controller *has* a database helper and a notification service (Spring injects them).
- **Inheritance / template method:** a JWT filter extends Spring's request filter and fills in authentication logic.
- **Layered design:** HTTP controllers, reusable services (security, commission, uploads), and a data-access class in front of the database.
- **Polymorphism:** Spring Security and exception handlers treat different error/filter types through common contracts.

6. Expected Outcome and Lab Contribution

At the end of the lab, the group will demonstrate a running system: register/login, book a service, change job status, submit a review, and demonstrate private verification and admin control. The code will be used to explain OOP in a real project—how objects share work, how roles restrict actions, and how data is saved safely. This proposal asks for faculty approval to complete WorkSure as our section A group project for Advanced Object Oriented Programming Language Lab.

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Submitted for evaluation and approval to the course faculty, Section A.